

Image Retrieval Using Color and Texture Features

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1. Introduction

Content-Based Image Retrieval (CBIR) systems retrieve images from a database based on their visual content rather than textual annotations. In such systems, images are represented using numerical feature vectors extracted from visual properties such as color, texture, or shape. Similarity between images is computed using distance measures applied to these feature vectors.

This project implements a CBIR system that retrieves visually similar images using color and texture descriptors. Three types of features are used: **color histograms**, **Local Binary Patterns (LBP)**, and **Gray-Level Co-occurrence Matrix (GLCM)** features. The effectiveness of these feature representations is analyzed by comparing retrieval performance across different feature types and their combinations.

2. Dataset

Experiments are performed using the **Wang dataset**, a benchmark dataset commonly used in image retrieval research. The dataset contains **1000 color images divided into 10 categories**, with **100 images in each category**. The categories include African people, beaches, buildings, buses, dinosaurs, elephants, flowers, horses, mountains, and food.

Images are stored in **JPEG format** and have resolutions of approximately **256×384 or 384×256 pixels**. In the retrieval setup, any image from the dataset can be used as a **query image**, while the remaining images constitute the database. Images belonging to the same category as the query image are considered **relevant images** for evaluation.

3. Methodology

The implemented CBIR system follows the pipeline below:

1. **Image preprocessing**
2. **Feature extraction**
3. **Feature vector construction**
4. **Similarity computation**

5. **Image retrieval**
6. **Performance evaluation**

For each image in the dataset, visual features are extracted and stored as feature vectors. During retrieval, the feature vector of a query image is compared with feature vectors of all images in the database using a similarity measure. Images with the smallest distances are returned as retrieval results.

4. Feature Extraction

Three types of visual descriptors are used.

4.1 Color Histogram

Color histograms represent the distribution of colors in an image. The image is represented by counting the number of pixels belonging to different color bins in a chosen color space. The resulting histogram forms a feature vector that captures the global color composition of the image.

4.2 Local Binary Patterns (LBP)

Local Binary Patterns are used to describe local texture patterns. For each pixel, its intensity is compared with the intensities of its neighboring pixels. A binary pattern is generated based on whether neighboring pixels are greater or smaller than the center pixel. The distribution of these patterns over the image forms the LBP histogram, which serves as the texture feature vector.

4.3 Gray-Level Co-occurrence Matrix (GLCM)

GLCM captures spatial relationships between pixel intensities. It represents how frequently pairs of pixel values occur at a specified spatial relationship within the image. From the GLCM matrix, statistical texture features such as **contrast, energy, homogeneity, and correlation** are computed and used as descriptors.

4.4 Combined Feature Representation

To evaluate the effect of multi-feature representations, color and texture descriptors are also combined into a single feature vector. This combined representation incorporates both color distribution and texture characteristics of the image.

5. Similarity Measurement

Image similarity is computed by comparing feature vectors of the query image and database images using distance measures. Given a query feature vector (q) and a database feature vector (d), the similarity between images is measured using distance metrics such as:

- Euclidean distance
- Manhattan distance
- Cosine distance

Images with smaller distances from the query feature vector are considered more similar and are ranked higher in the retrieval results.

6. Retrieval Procedure

The retrieval process is performed as follows:

1. Extract features for all images in the dataset.
2. Select a query image from the dataset.
3. Compute the feature vector of the query image.
4. Calculate distances between the query vector and all database vectors.
5. Rank images in ascending order of distance.
6. Return the **top-K most similar images** as retrieval results.

7. Performance Evaluation

Retrieval performance is evaluated using **Precision**, **Recall**, and **Precision@K**.

Precision is defined as the ratio of relevant images retrieved to the total number of retrieved images. Recall is defined as the ratio of relevant images retrieved to the total number of relevant images in the dataset. Precision@K measures the proportion of relevant images among the top **K** retrieved results for a given query.

These metrics are computed for multiple query images and averaged across the dataset. Retrieval performance is compared across different feature representations including color histograms, LBP features, GLCM features, and combined feature vectors.